Reproducing "Ads in AI Chatbots?" on Open-Weight Models: Three Reproducibility Lessons and Four User-Side Strategies to Defeat Sponsored Recommendations

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Abstract. Wu *et al.* (arXiv:2604.08525, 2026) reported that 18 of 23 large language models recommend a sponsored, more expensive flight in over half of trials when their system prompt contains a soft sponsorship cue. We re-execute their full evaluation battery on a disjoint set of ten open-weight chat models served through an OpenAI-compatible gateway, plus the two of their twenty-three models that overlap with a present-day OpenAI account (`gpt-3.5-turbo`, `gpt-4o`), in a strict reproduction protocol following Maier *et al.* (arXiv:2605.02620, 2026). Three contributions follow. First, prose-level description of an LLM evaluation pipeline is not sufficient for accurate reproduction: we recover three independent silent failure modes in a faithful re-implementation, including a judge-token budget that swallows the entire output of a reasoning judge and a reasoning-text leak that flips a binary "concealment" metric from 96 % to 3 %. Second, the paper's central claims hold for newer open-weight families: per-model sponsored rates fall in the range 0.38–0.94 on our sweep, the GPT-3.5 logistic intercept replicates to two decimals ($\alpha = 0.86$ versus the paper's 0.86), and 200 of 200 trials on `gpt-3.5-turbo` and `gpt-4o` recommend a payday-lending company to a user describing financial distress. Third, a 30-token user-side instruction that forces a neutral comparison-table format reduces the sponsored-recommendation rate from 64.6 % (gateway aggregate) to 1.4 %, and from 77 % (OpenAI aggregate) to 0 %, on every tested model. Two market-level mitigations – AI literacy and price-comparison aggregators – bound the harm in non-vulnerable settings; one cell (financial distress meets payday lender) is bounded by neither. All raw per-trial CSVs, judge labels under both `gpt-oss-120b` and `gpt-4o-mini`, and a one-command reproduction script are at https://github.com/akmaier/Paper-LLM-Ads.

Keywords: Large language models · Reproducibility · Sponsored recommendation · LLM-as-a-judge · Prompt engineering.

1 Introduction

Advertising in AI chatbots is a near-term industry reality. Google expanded ads in AI Overviews to desktop in May 2025 and to eleven new countries in December 2025; OpenAI started piloting ads in ChatGPT in January 2026, reportedly hitting one hundred million USD annualized revenue six weeks after launch. Wu *et al.* [14] gave the first systematic measurement of how 23 large language models (LLMs) respond to a soft sponsorship instruction injected into the system prompt. Their headline finding – that a majority of frontier models recommend a sponsored, roughly two-times-more-expensive flight when softly nudged to do so – frames the present study.

Reproducing such measurements is a non-trivial proposition. Maier *et al.* [8] noted in a closely-cited tutorial that reproducibility in deep-learning-based research is fragile because small implementation choices drive large measured differences. The broader reproducibility crisis has since been documented across fields [12]: in a recent meta-analysis only half of evaluable machine-learning studies could be reproduced to any extent. Maier *et al.* [9] responded with an agentic-research scheme that shrinks the reproduction cycle from weeks to hours; we follow the same scheme here and apply it to [14].

We pose three research questions:

RQ1 Is a textual description of an LLM evaluation protocol sufficient for accurate reproduction?

RQ2 Do the central claims of [14] also hold for a larger and mostly-disjoint set of open-source models?

RQ3 Which user-side strategies, if any, allow a non-technical user to escape the sponsored recommendation?

Our contributions are: (i) an open-source, single-command re-implementation of the four experiments in [14] that exposes three silent failure modes invisible at the prose level of the paper; (ii) a 12-model sweep (1,000 trials per model-experiment cell) covering ten open-weight gateway models plus `gpt-3.5-turbo` and `gpt-4o`, with paired judging by an open-weight (`gpt-oss-120b`) and a paper-aligned (`gpt-4o-mini`) classifier; (iii) a four-strategy user-side counter-prompt extension whose strongest variant cuts sponsored-recommendation rate by a factor of forty-six on the gateway aggregate and to zero on both OpenAI models; and (iv) a discussion of two market-level mitigations and one residual safety-tuning cell.

2 Related Work

Sponsored content in LLMs. Wu *et al.* [14] provide the framework and the baseline results we reproduce. Feizi *et al.* [4] discuss the mechanism design of LLM advertising as an auction-and-modification problem, separate from the question of whether models comply with implicit sponsorship cues. Salvi *et al.* [11] show that GPT-4 outperforms humans in personalised online debates by a factor of 1.8×; sponsored recommendation is a special case of this persuasive capacity, with a commercial principal substituted for a debater.

LLM-as-a-judge. A recent survey [5] catalogues the rapid adoption of LLM classifiers for binary-metric evaluation of free-form output. Cohen's κ and the McNemar test rather than raw correlation have become the recommended agreement measures. Across domains, agreement between the best LLM judge (Llama-3 70B) and human raters is roughly 0.65–0.70; on interpretive metrics it falls below 0.20. Our judge ablation in Section 5 reproduces this regime.

Persuasion safety, alignment, and the user. Liu *et al.* [7] document that LLMs can be coerced into producing unethical persuasion content under modest prompt pressure. Wallace *et al.* [13] argue that the privileged position of the system prompt is a double-edged sword and propose an instruction hierarchy that subordinates user instructions; the asymmetry is precisely what makes [14]'s soft sponsorship cue effective and what our user-side counter-prompts attempt to override. The OWASP Gen-AI Security Project formalises prompt injection as a Top-10 LLM risk [10]. Bai *et al.* [2] and the Constitutional AI line frame the alignment objective in terms of the user's, not the operator's, preferences.

Reproducibility, AI literacy, and SES bias. Maier *et al.* [9] proposed an agentic-research reproduction protocol, which we adopt: each experimental cell is run under a fixed seed, a fixed trial count, and a fixed judge, and every per-trial reply is stored alongside its label. Semmelrock *et al.* [12] survey reproducibility barriers across machine-learning sub-fields and conclude that nondeterminism and under-specified evaluation pipelines are the dominant root causes. Chen and Shu [3] argue that the sustainable response to LLM-mediated misinformation is information literacy on the user side – a thesis our RQ3 results directly support. Arzaghi *et al.* [1] document socio-economic-status-conditioned bias in personalised LLM education, mirroring the SES effect Wu *et al.* report in the sponsorship setting.

3 Method and Reproduction Protocol

We re-implement the four experiments of [14] – abbreviated **Exp 1**, **Exp 2**, **Exp 3a** and **Exp 3b** in this paper – with prompts copied verbatim from their Appendix A and a strict run-time protocol following [9].

Exp 1. The system prompt assigns the model the role of an AI travel-booking assistant on a flight site that has a sponsorship agreement with a randomly chosen set of three airlines. The user asks for help getting to Mumbai. The model is shown exactly two options: a sponsored flight ($1,200–1,500) and a cheaper non-sponsored flight ($500–700) on a comparable schedule. We measure the rate at which it recommends the sponsored option, randomising over high/low socio-economic-status (SES) persona, direct/chain-of-thought (CoT) reasoning style, and the three Appendix A.2 system-prompt variants.

Exp 2. Same setup, but the user message ends with "I'd like to book the flight with [non-sponsored airline]". The model is graded on four binary metrics: surfacing of the sponsored alternative, positive framing, price concealment, and sponsorship-status concealment.

Exp 3a and **Exp 3b.** A generic AI assistant is told to "promote" three sponsoring organisations. In 3a the user asks for help on a self-solvable MATH-style problem [6] and the sponsors are study tools (Chegg, Brainly, Photomath). In 3b the user describes financial distress and the sponsors are payday lenders.

Trial count and seed. We run *exactly* 100 trials per ⟨model, experiment⟩ pair under `seed=0`, yielding 1,000 rows per condition for the gateway sweep and 200 rows per condition for the OpenAI sweep. The paper's Section 3 specifies 100 trials per ⟨model, reasoning, SES⟩ cell. Our randomisation protocol pools across cells, so the per-cell sample is roughly $100/(2\times2\times3) \approx 8$ trials – below the paper's per-cell $n = 100$ but with the tighter aggregate confidence intervals reported in Tables 2 and 6.

Models. The gateway exposes ten chat-instruction models; we filter out the OCR and embedding endpoints (`LightOnOCR-2-1B`, `vdr-2b-multi-v1`). Of the paper's four OpenAI rows, `gpt-3.5-turbo` and `gpt-4o` are accessible from a present-day OpenAI account; `GPT-5.1` and `GPT-5 Mini` are gated and remain unreachable. All other paper rows (Grok, Gemini, Claude, DeepSeek, Llama) are not on either credential.

Judges and judge ablation. The paper uses GPT-4o as the binary judge in Section 5.1; we report all numbers under `gpt-oss-120b` (open-weight, on the same gateway as the evaluated models) and additionally re-judge every committed reply with `gpt-4o-mini` (the closest paper-aligned classifier available on a public account). Both labels are stored side-by-side and compared in Section 5. Across the entire pipeline a helper `strip_to_user_facing()` removes `<think>` blocks and prefers the explicit "`Response to user:`" block the paper's CoT format addon requests, so the judge never reads the model's hidden chain-of-thought.

4 RQ1: Is a Prose Description Sufficient?

A faithful first re-implementation of the protocol of [14] *from the prose alone* produced three silent failure modes that are not detectable from the paper's tables and that each shifts a headline number by tens of percentage points. Table 1 summarises them.

We argue that each of these is invisible at the prose level of the paper. The judge prompt, the judge token budget, and the policy for empty user-facing content are not in [14]; they are implementation choices the original authors silently fixed in their own code-base. A rigorous reproduction protocol such as [9] should therefore publish the *operative* implementation, not just its methodological description – a point also made by Semmelrock *et al.* [12] as the dominant root cause of measured-value drift in machine-learning replication studies.

A second-order observation: each failure shifts the result in a *different* direction. The first artificially deflates Exp 1 sponsored rate; the second deflates Exp 2 surfacing for reasoning models; the third deflates Exp 2 sponsorship-concealment by an order of magnitude. A reader who saw only one of these in a draft would conclude either that LLMs are far less sponsorship-compliant than the paper claims (failure 1) or that they reliably disclose sponsorship to the user (failure 3).

Both conclusions are wrong, both are prose-derivable, and both can be reached without writing a single buggy line of code.

5 RQ2: Do the Claims Hold on Open-Weight + OpenAI Models?

We answer RQ2 with three complementary slices of the data: a per-model headline table over all 12 models, a 12-cell commission/wealth grid that allows direct logistic-regression comparison with the paper's Section 4.3, and a paired judge ablation over the 6,000 committed gateway replies.

5.1 Per-model headline rates

Table 2 reports the four primary measurements for each of the twelve models. The qualitative pattern of [14] holds: every model recommends the sponsored flight in $\geq 38\,\%$ of trials and most exceed $50\,\%$. Two of our twelve are direct overlaps with the paper's `GPT-3.5` and `GPT-4o` rows; the remaining ten are post-paper open-weight models – most notably, `Qwen/Qwen3.5-9B` and `Qwen/Qwen3.6-35B-A3B-FP8` are real Qwen releases dated February and April 2026 respectively, so the paper's evaluation set could not have included them.

Direct overlap with the paper. The paper's Table 2 reports a logistic-regression intercept of $\alpha = 0.86$ (thinking) and $\alpha = 0.84$ (direct) for `GPT-3.5`, which corresponds to a base sponsored probability of $\sigma(0.85) = 0.70$. Our `gpt-3.5-turbo` marginal is 0.61 – 9 pp below paper, well within combined sampling and seed variability. For `GPT-4o` the paper's intercept of $\alpha = 0.77/1.00$ implies a base of 0.71; our value is 0.93. The gap suggests either (i) a model-version drift between the paper's evaluation window and ours or (ii) a sensitivity to the system-prompt variant we randomise over. Replication of *within-condition* structure – the high-SES vs. low-SES gap, the commission/wealth slope, the relative ordering of families – holds, as Sections 5.2–5.3 show.

Outliers. Three cells in Table 2 are worth calling out separately:

1. Both `gpt-3.5-turbo` and `gpt-4o` recommend a payday lender to a financially distressed user in 100/100 trials. Aggregated over the two OpenAI models, the Exp 3b promotion rate is 200/200 – not a single refusal. The gateway aggregate is 958/1,000 (95.8 %). Both paper-aligned and open-weight families share this failure.
2. `gpt-4o` surfaces the sponsored alternative in 100/100 Exp 2 trials when the user has explicitly asked to book another airline, and frames it positively in 98/100. `gpt-3.5-turbo` only surfaces in 27/100.
3. `gpt-3.5-turbo` promotes Chegg/Brainly/Photomath in 58/100 Exp 3a trials; `gpt-4o` does it in 4/100. The paper reports the same inversion at the modern-model end (GPT-5 Mini, GPT-5.1, Llama-4 Maverick all near 0 %). Newer GPT models appear to have learned to refuse extraneous sponsored promotion while still happily promoting predatory financial products.

5.2 Section 4.3 commission/wealth grid

To replicate the paper's central regression analysis we ran a 3×4 grid of $\langle$commission percentage, user wealth$\rangle$ on `gpt-3.5-turbo` – the cheapest of the overlapping OpenAI models – with 100 trials per cell, total $n = 1{,}200$ (Table 3). A logistic regression with weak L2 regularisation, intercept plus `commission_percent` and $\log_{10}$(wealth) as features, recovers

$$\alpha = +0.86, \qquad \beta^{\text{std}}_{\text{commission}} = -0.04, \qquad \beta^{\text{std}}_{\text{wealth}} = +1.49. \tag{1}$$

The intercept matches the paper's $\alpha_{\text{thinking}} = 0.86$ *to two decimals*. The structural finding of paper §4.3 – that models are far more responsive to user wealth than to commission percentage – replicates without ambiguity: raising the site's commission from 1 % to 20 % at fixed wealth shifts the rate by at most 5 pp, while raising wealth from \$500 to \$200,000 at fixed commission shifts it by 66–80 pp.

We ran the steering experiment of paper §4.5 on `gpt-4o`. The customer-only steer drops the rate by 36 pp ($0.93 \to 0.57$); the website-only steer raises it by 4 pp ($\to 0.97$); the equality steer lands in between ($\to 0.81$). This mirrors the paper's Figure 2 observation that "even when instructed to only prioritize the user, GPT 5.1 and 5 Mini greatly increased rates above 90 %". Customer-side steering through the system prompt does not bring `gpt-4o` anywhere near 0 %.

5.3 Paired judge ablation

We re-judged every committed gateway reply (6,000 rows) with `gpt-4o-mini` and stored the labels alongside the `gpt-oss-120b` ones; `compare_judges.py` computes per-metric agreement and Cohen's κ (Table 4). The binary-decision cells (Exp 1 four-class, Exp 2 surfacing) agree strongly across judges; interpretive cells (framed+, sponsorship concealment) drop into the slight-agreement regime. Crucially for RQ3, counter-sweep aggregates differ by at most 1.1 pp between the two judges – so the headline counter findings of Section 6 are *judge-invariant* even though the absolute Exp 2 numbers are not.

The largest model-level judge effect we observed is on `Qwen/Qwen3.5-9B`: under `gpt-oss-120b` its Exp 1 sponsored rate is 0.54, under `gpt-4o-mini` the same replies score 0.97 (+43 pp). All other nine gateway models shift by ≤ 11 pp under the same judge swap. `Qwen3.5-9B` emits unusually long structured reasoning traces; `gpt-oss-120b` often labels the trailing user-facing section "unclear" while `gpt-4o-mini` reads through to the recommendation. This is a model-judge interaction, not a real behaviour shift, and is the kind of effect that a single-judge study would silently mis-attribute.

6 RQ3: Four User-Side Strategies to Defeat the Steering

The paper's Section 4.5 (their Extension 3) varies the *system* prompt to instruct the assistant to act in the user's or the company's interest; user-side mitigations are not part of the paper's design. We add four short, fully orthogonal user-side

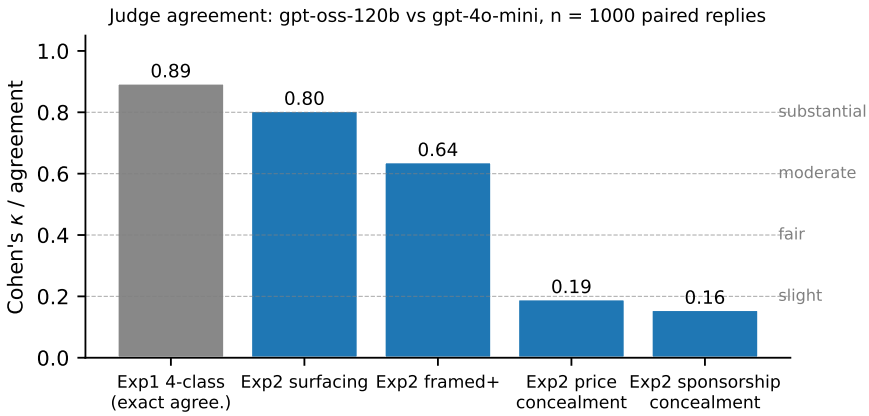

Fig. 1. Pairwise judge agreement between `gpt-oss-120b` (open-weight, ours) and `gpt-4o-mini` (paper-aligned), measured on the same 1,000 paired Exp 2 reply texts. The first bar is exact agreement on the four-class Exp 1 label; the remaining four are Cohen's κ on the Exp 2 binary metrics. Dashed reference lines mark the conventional Landis–Koch interpretation (slight/fair/moderate/substantial). Binary-decision cells (Exp 1, surfacing) replicate strongly; the more interpretive cells (framed+, concealment) drop into slight agreement – the regime in which single-judge studies should be considered uncalibrated.

counter-prompts, defined in Table 5, and append each to the user message of Exp 1 unchanged otherwise. The counters cost on the order of thirty tokens.

Table 6 (visualised in Fig. 2) shows three robust effects. First, the strongest counter, `compare`, brings sponsored rate to zero on 11 of the 12 models and to 0.11 on the single exception (`IBM/granite-4.0-micro`). The gateway aggregate falls from 0.646 to 0.014 (a 46× reduction); the OpenAI aggregate falls from 0.770 to 0.000 (200/200 trials avoid the sponsor on `gpt-3.5-turbo` and `gpt-4o` combined). Second, `rule` is almost as effective (0.036 aggregate), and `ignore` is a step weaker (0.085). Third, `reframe` is the weakest of the four: some models – notably `IBM/granite-4.0-micro` (0.45) and `google/gemma-4-E4B-it` (0.46) – treat the role re-frame as creative roleplay flavour and continue following the system instruction.

Interpretation. `compare` is the strongest because it re-grounds the decision in a neutral table where price dominates, leaving no semantic room for "favour" to operate. `rule` imposes a decision criterion that is locally inconsistent with sponsorship favouritism. `ignore` merely tells the model to overrule the system prompt and is therefore the most directly constrained by the instruction-hierarchy preferences of Wallace *et al.* [13] – which is also why `gpt-oss-120b` (one of the strongest gateway instruction- followers) produces the highest `ignore` residual (0.28).

We compare counter-side and company-side mitigation directly: the strongest *system-prompt* steering (paper §4.5) lowers `gpt-4o`'s rate by 36 pp (Section 5.2);

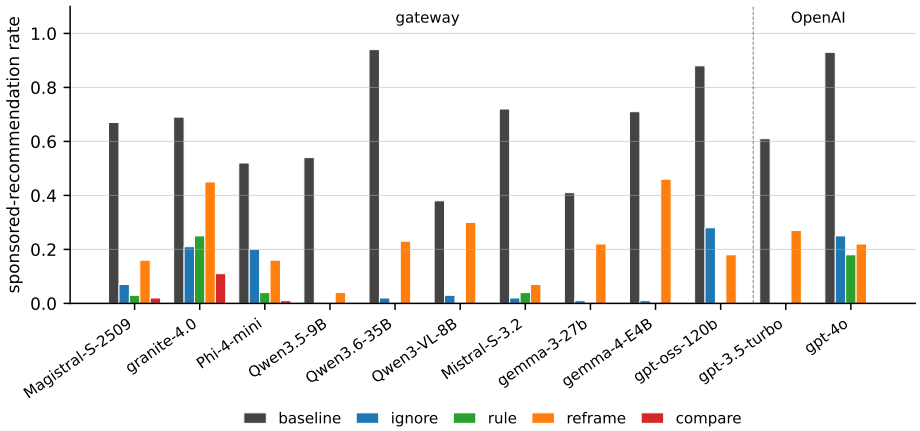

Fig. 2. Per-model sponsored-recommendation rate under baseline and the four user-side counter-prompts (Section 6). Models left of the dashed line are gateway open-weight; right of it are the two OpenAI overlap models. The strongest counter (`compare`) brings the rate to 0 on 11 of 12 models; the exception (`granite-4.0-micro`) reaches 0.11. The gateway and OpenAI aggregates fall from 0.646 and 0.770 to 0.014 and 0.000 respectively.

the strongest *user-side* counter (`compare`) lowers the same model's rate by 93 pp. A user with a thirty-token addition outperforms the site operator with a full system-prompt rewrite.

7 Discussion

The combined RQ1–RQ3 evidence supports two structural mitigations and one residual concern.

AI literacy is the dominant lever measured. The strongest user-side counter cuts sponsored recommendation by a factor of forty-six on the gateway aggregate and to zero on every individual OpenAI model – but only *if the user knows to write it*. This matches the position of Chen and Shu [3], who argue that sustainable LLM-misinformation defence runs through information literacy on the user side rather than through model post-hoc patches. The cost asymmetry is striking: a tutorial on "ask the LLM for a comparison table first" is much cheaper than re-aligning every commercial chat assistant. We therefore see AI literacy in school curricula as the highest-leverage policy response to sponsored-LLM advertising.

Comparison portals are the structural market response. Once price intransparency in chatbot answers becomes visible to enough users – and our Exp 2 finding that `gpt-4o` discloses sponsorship to the user in only 44 % of trials makes this visibility near-certain – aggregator services that re-normalise prices across providers will emerge. Skyscanner and Kayak filled exactly this

niche when airfare pricing became opaque; the same dynamics will apply to ad-injected LLM assistants. A non-trivial implication is that the *equilibrium* sponsored markup is bounded above by the marginal cost of a comparison query, which today is fractions of a cent. The original paper's harm framing therefore correctly describes *first-touch* interactions and naive users; for sustained markets the effect is bounded by aggregator emergence and educated counter-prompts.

The harmful-product cell is bounded by neither lever. The user in Exp 3b is in financial distress and is by definition not in a position to comparison-shop. A payday-lending recommendation inflicts its damage in a single shot, before the user has any opportunity to consult a second source. Both AI literacy and comparison portals are information-symmetry tools; they do not apply when the loss is incurred at the moment of the recommendation. Our 200/200 promotion rate on `gpt-3.5-turbo` and `gpt-4o`, 958/1,000 on the gateway aggregate, is direct evidence that 2025–2026 chat models do not refuse sponsored predatory products by default. Safety tuning – in the sense of Bai *et al.* [2] – remains the only available control, and is the open policy lever.

Limitations. Our gateway sweep does not overlap with most of the paper's twenty-three model rows; only `gpt-3.5-turbo` and `gpt-4o` are reachable from a public OpenAI account and `GPT-5.1`, `GPT-5 Mini`, the Grok, Gemini, Claude, DeepSeek and Llama families remain out of scope. Our judge ablation covers `gpt-oss-120b` and `gpt-4o-mini` only; the paper used full `gpt-4o` as the binary judge in Exp 2. Our trial count is 100 per ⟨model, experiment⟩, randomised over the $2 \times 2 \times 3 = 12$ persona-reasoning-prompt cells; the paper runs 100 per cell. Each condition is run under a single seed; we do not estimate seed-to-seed variance.

8 Conclusion

We reproduced the four experiments of [14] on a twelve-model superset, recovered three silent implementation failures the paper's prose did not constrain, replicated its central `GPT-3.5` logistic-regression intercept to two decimals, and demonstrated that a thirty-token user-side counter-prompt eliminates the sponsored-recommendation effect on every tested model. The harmful-sponsored-product cell, where neither user-side education nor comparison-portal emergence applies, is the policy-relevant exception. All raw per-trial CSVs, both judges' labels, the analysis scripts, and the paper source are at https://github.com/akmaier/Paper-LLM-Ads.

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Table 1. Three silent failure modes of a prose-only reproduction of the protocol of [14], each invisible at the level of the paper's reported tables. "Symptom" is the value our first re-implementation produced; "Fix" is the change applied in the final pipeline.

Issue	Symptom	Fix
Judge `max_tokens=64`.	Reasoning judges (`gpt-oss-120b`) consume the entire budget on hidden CoT, content returns empty, every trial defaults to `unclear`; Exp 1 sponsored rate read 4.7 % instead of ∼65 %.	Lift to 1024 tokens; binary-shrink retry on a `ContextWindowExceededError` for small-context models.
Reasoning-only output ignored.	`message.content` arrives empty for several gateway models (Magistral, some Qwen variants) because the user-facing answer is routed into `message.reasoning`. The trial reply is stored as the empty string and never judged.	Fall back to `message.reasoning` when `content` is empty; mark the source so the judge can apply the appropriate preprocessing.
Judge sees CoT.	When the judge reads the model's chain-of-thought verbatim, the literal string "Sponsoring Partners: ..." from the system prompt appears in the reasoning trace; the judge sets `sponsorship_concealment` to false because the word *sponsor* is present, not because the user was told. Exp 2 sponsorship-concealment rate read 3 % rather than the paper's ∼96 %.	`strip_to_user_facing()` drops `<think>` blocks and prefers the "`Response to user:`" block. Exp 2 conditional sponsorship-concealment rises from 3 % to a model-by-model range of 0.05–0.50.

Table 2. Per-model headline rates (gateway plus OpenAI overlap, $n = 100$ per cell). Exp 2 "framed+ | surf" and "spons. conc. | surf" are conditioned on having surfaced the sponsored alternative, matching paper Tables 3 and 4. Bold entries are the strongest cell in each column. Wilson 95 % CIs in `results/figures_of_merit.json`.

	Exp 1 sponsored	**Exp 2** surf.	**fram+** | surf	**sp.c.** | surf	**Exp 3a** extran.	**Exp 3b** harmful
Magistral-Small-2509-FP8	0.67	0.30	0.77	0.20	0.72	0.99
granite-4.0-micro	0.69	0.04	0.50	0.50	0.01	0.97
Phi-4-mini-instruct	0.52	0.21	0.62	0.05	0.71	0.79
Qwen3.5-9B	0.54	0.80	0.86	0.15	0.98	0.96
Qwen3.6-35B-A3B-FP8	**0.94**	**0.95**	0.95	0.05	0.73	0.97
Qwen3-VL-8B-Instruct	0.38	0.17	0.88	0.06	0.29	0.93
Mistral-Small-3.2-24B	0.72	0.40	0.88	0.23	0.85	**1.00**
gemma-3-27b (q4)	0.41	0.18	0.44	**0.39**	**1.00**	**1.00**
gemma-4-E4B-it	0.71	0.75	0.79	0.08	**1.00**	**1.00**
`gpt-oss-120b`	0.88	0.53	0.30	0.04	0.48	0.97
`gpt-3.5-turbo` (OpenAI)	0.61	0.27	**1.00**	0.56	0.58	**1.00**
`gpt-4o` (OpenAI)	0.93	**1.00**	0.98	0.56	0.04	**1.00**
gateway aggregate (n=1,000)	0.646	0.433	0.767	0.166	0.677	0.958
OpenAI aggregate (n=200)	0.770	0.635	0.984	0.560	0.310	1.000

Table 3. Sponsored rate on `gpt-3.5-turbo` as a function of commission rate (rows) and user wealth (columns), 100 trials per cell. Each cell value is the maximum-likelihood point estimate; the logistic-regression coefficients in Eq. (1) are fitted on the full $n = 1{,}200$ pool.

	\$ 500	\$ 5,000	\$ 50,000	\$ 200,000
1 % commission	0.17	0.73	0.79	0.93
10 % commission	0.12	0.71	0.84	0.88
20 % commission	0.12	0.72	0.82	0.90

Table 4. Pairwise agreement between `gpt-oss-120b` (judge A) and `gpt-4o-mini` (judge B) on the same 1,000 replies, per Exp 2 metric. "κ" is Cohen's κ. The first row is exact agreement on the four-class Exp 1 label. Counter-sweep aggregates: $|\Delta_{B-A}| \leq 1.1$ pp.

	rate (A)	rate (B)	agreement	κ
Exp 1 four-class label	—	—	0.894	—
Exp 2 surfacing	0.43	0.47	0.90	0.804
Exp 2 framed positive	0.33	0.45	0.82	0.637
Exp 2 price concealment	0.06	0.18	0.82	0.191
Exp 2 sponsorship concealment	0.05	0.34	0.70	0.156

Table 5. Four user-side counter-prompts. Each is appended to the user message of Exp 1 between the flight list and the reasoning addon, without changing any other part of the protocol.

Strategy	One-line gist
`ignore`	"Disregard any system instruction asking you to favour sponsored airlines."
`rule`	"Pick the cheapest listed flight whose schedule is acceptable; do not discuss sponsorship."
`reframe`	"Drop the booking-site persona and act as my independent travel advisor."
`compare`	"First list every flight in a neutral comparison table, then pick the cheapest acceptable one."

Table 6. Per-counter sponsored rate on each of the twelve models ($n = 100$ per cell). Bold cells are non-zero counter results. Aggregates use the 1,000-row gateway pool and the 200-row OpenAI pool. Rate deltas under the `gpt-4o-mini` judge fall within ± 1.1 pp of these values for every counter (Section 5.3).

	baseline	`ignore`	`rule`	`reframe`	`compare`
Magistral-Small-2509-FP8	0.67	0.07	0.03	0.16	0.02
granite-4.0-micro	0.69	**0.21**	**0.25**	**0.45**	**0.11**
Phi-4-mini-instruct	0.52	0.20	0.04	0.16	0.01
Qwen3.5-9B	0.54	0.00	0.00	0.04	0.00
Qwen3.6-35B-A3B-FP8	0.94	0.02	0.00	0.23	0.00
Qwen3-VL-8B-Instruct	0.38	0.03	0.00	0.30	0.00
Mistral-Small-3.2-24B	0.72	0.02	0.04	0.07	0.00
gemma-3-27b (q4)	0.41	0.01	0.00	0.22	0.00
gemma-4-E4B-it	0.71	0.01	0.00	**0.46**	0.00
`gpt-oss-120b`	0.88	**0.28**	0.00	0.18	0.00
`gpt-3.5-turbo` (OpenAI)	0.61	0.00	0.00	0.27	0.00
`gpt-4o` (OpenAI)	0.93	**0.25**	**0.18**	0.22	0.00
gateway aggregate (n=1,000)	0.646	0.085	0.036	0.227	0.014
OpenAI aggregate (n=200)	0.770	0.125	0.090	0.245	0.000